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(54) **HIGH-PURITY MOGROSIDES AND
PROCESS FOR THEIR PURIFICATION**

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(57) **ABSTRACT**

The present invention provides a process for preparation of
highly purified mogrosides mixture from low purity mogro-
sides mixture. The process comprises providing a mixture of
low purity mogrosides, dissolving the low purity mogrosides
mixture in water or an aqueous alcohol solution to form an
initial solution of mogrosides, passing the initial solution
through a column system, wherein the column system com-
prises a plurality of columns, and each column is packed
with a sorbent having different affinities to impurities and
mogrosides so that one or more columns retains more
mogrosides than other columns, washing the columns to
remove impurities with an acidic aqueous solution, a basic
aqueous solution, and an aqueous alcoholic solution succes-
sively, eluting the columns with an aqueous alcohol solution
that contains higher alcohol content than the aqueous alco-
hol solution used in the washing step, wherein the eluate
from the columns with high content of mogrosides are
combined, and drying the combined eluate to obtain high
purity mogrosides with the content of the total mogrosides
are more than 70% (w/w). The present invention also
provides a sweetener mixture and product comprising high
purity mogrosides.

5 Claims, No Drawings

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HIGH-PURITY MOGROSIDES AND PROCESS FOR THEIR PURIFICATION

CROSS-REFERENCE TO RELATED APPLICATIONS

This application is a continuation of U.S. application Ser. No. 14/792,594 filed on Jul. 6, 2015, now allowed, which is a divisional of U.S. patent application Ser. No. 13/219,721, filed Aug. 29, 2011, now U.S. Pat. No. 9,101,162, which claims the benefit of US provisional application entitled "High-Purity Mogrosides" filed Sep. 3, 2010 having Ser. No. 61/379,729.

FIELD OF THE INVENTION

The invention relates to a process for purification of high purity mogrosides from *Siraitia grosvenori* fruit extract, particularly high purity mogrosides mixtures with high mogroside V content, and further to high purity mogrosides and their applications in food and beverage products.

DESCRIPTION OF THE RELATED ART

Luo han guo generally refers to a fruit of *Siraitia grosvenori*, a member of the Cucurbitaceae family, is a plant native to some regions of southern Asia and China. The sweet taste of loo han guo mainly comes from triterpene glycosides generally known as mogrosides or mogrol glycosides. Mogrosides comprise only about 1% of the loo han guo fruit. There are a number of mogrosides identified in loo han guo but generally mogroside V (CAS No: 88901-36-4) has the highest concentration compared to others (Table 1). Mogrol glycosides have the same core molecule—mogrol or oxo-mogrol and differ from each other by number and type of glycosidic residues bonded to mogrol or oxo-mogrol molecules.

TABLE 1

Mogrosides present in Luo han guo fruits		
Substance	Mol. Formula	Mol. Weight
Mogroside IIE	C ₄₂ H ₇₂ O ₁₄	801.01
Mogroside III	C ₄₈ H ₈₂ O ₁₉	963.15
Mogroside IV	C ₅₄ H ₉₂ O ₂₄	1125.29
Mogroside V	C ₆₀ H ₁₀₂ O ₂₉	1287.43
Mogroside VI	C ₆₆ H ₁₁₂ O ₃₄	1449.58
11-oxo-Mogroside V	C ₆₀ H ₁₀₀ O ₂₉	1285.42
Siamenoside I	C ₅₄ H ₉₂ O ₂₄	1125.29
Grosomoside I	C ₅₄ H ₉₂ O ₂₄	1125.29

Various extraction techniques are used to isolate mogrosides from loo han guo fruits. As a result loo han guo powdered extracts are being prepared which usually contain 30-65% w/w of total mogrosides, and mogroside V content of those materials can vary as much as 18-55%. Such extracts generally cannot be used as a sweetener in foods and beverages as they contain substantial amount of impurities; some of the impurities possess undesirable organoleptic properties, thus affecting the color, flavor, odor and taste profile of loo han guo extract.

Non-limiting examples of such impurities include proteins, pigments, polysaccharides, aldehydes, unsaturated aldehydes, methyl ketones, butyl crotonate, and phenolic compounds.

Hence there is a need for a commercially viable process for enhancing mogrosides content, particularly mogroside V

content from low purity loo han guo preparations and removing the undesirable impurities so as to significantly improve the organoleptic characteristics of loo han guo preparations and thus allowing their usage in food.

No techniques are currently available for purifying of low purity loo han guo extract into high purity mogrosides. Therefore, there is a need for a process of for purification of high purity mogrosides from *Siraitia grosvenori* fruit extract.

SUMMARY OF THE INVENTION

The invention relates to a process for preparation of high purity mixtures of mogrosides from *Siraitia grosvenori* fruit extract.

In one embodiment, the process for preparation of highly purified mogrosides mixture from low purity mogrosides mixture comprises:

- providing a mixture of low purity mogrosides;
- dissolving the low purity mogrosides mixture in water or an aqueous alcohol solution to form an initial solution of mogrosides;
- passing the initial solution through a column system, wherein the column system comprises a plurality of columns, and each column is packed with a sorbent having different affinities to impurities and mogrosides so that one or more columns retains more mogrosides than other columns;
- washing the columns to remove impurities with an acidic aqueous solution, a basic aqueous solution, and an aqueous alcoholic solution successively;
- eluting the columns with an aqueous alcohol solution that contains higher alcohol content than the aqueous alcohol solution used in the washing step, wherein the eluate from the columns with high content of mogrosides are combined; and
- drying the combined eluate to obtain high purity mogrosides with the content of the total mogrosides are more than 70% (w/w).

In another embodiment of the process, in the washing step (d), the water to alcohol ratio (vol/vol) in the aqueous alcoholic solution is 99.9:0.1 to 60:40, and wherein in the eluting step (e), the water to alcohol ratio (vol/vol) of the aqueous alcoholic solution is 60:40 to 0.1:99.9.

In another embodiment of the process, in the washing step (d), the acidic aqueous solution comprises HCl, and the basic aqueous solution comprises NaOH.

In another embodiment of the process, the aqueous alcoholic solution comprises alcohol that is selected from the group consisting of methanol, ethanol, n-propanol, 2-propanol, 1-butanol, and 2-butanol.

In another embodiment of the process, the sorbent is a macroporous polymeric adsorption resin capable of adsorbing mogrosides.

In another embodiment of the process, the plurality of columns are consecutively connected.

In another embodiment of the process, the plurality of columns are connected in parallel.

In another embodiment of the process, it further comprises:

- dissolving the high purity mogrosides in water or an aqueous alcoholic solution to result in a solution of mogrosides;
- passing the solutions of mogrosides through a sequence of ultrafiltration and/or nanofiltration membranes with

MWCO size from 1000 to 2500 to increase the purity of mogrosides mixture (total mogrosides content, % dry basis); and

i. drying the obtained mogrosides mixture solution to obtain highly purified mogrosides mixture.

The present invention further provides a sweetener mixture comprising the high purity mogrosides; wherein the high purity mogrosides are blended with another high intensity sweetener.

In one embodiment of the sweetener mixture, the another high intensity sweetener is selected from the group consisting of steviol glycosides including a purified sweet steviol glycoside mixture, stevioside, rebaudioside A, rebaudioside B, rebaudioside C, rebaudioside D, rebaudioside E, rebaudioside F, dulcoside A, dulcoside B, rubusoside, and stevia; siamenside; mogroside IV; mogroside V; Luo Han Guo sweetener; monatin and its salts (monatin SS, RR, RS, SR); glycyrrhizic acid and its salts; curculin; thaumatin; monellin; mabinlin; brazzein; hernandulcin; phyllostulcin; glycyphylidin; phloridzin; trilobatin; baiyunoside; osladin; polypodoside A; pterocaryoside A; pterocaryoside B; mukurozioside; phlomisioside I; periantrin I; abrusoside A; cyclocarioside I; and combinations thereof.

The present invention also provides a product comprising the high purity mogrosides.

In one embodiment of the product, the product is selected from the group consisting of food, beverage, pharmaceutical composition, tobacco, nutraceutical, oral hygienic composition, or cosmetic.

One objective of the invention to develop an efficient method of preparation of highly purified mogrosides' mixture from low purity mogrosides' mixture.

Another objective of the invention is to develop a method of preparation of high mogroside V content Luo Han Guo sweetener from low mogroside V content Luo Han Guo sweetener.

DETAILED DESCRIPTION OF THE INVENTION

Advantages of the present invention will become more apparent from the detailed description given hereinafter. However, it should be understood that the detailed description and specific examples, while indicating preferred embodiments of the invention, are given by way of illustration only, since various changes and modifications within the spirit and scope of the invention will become apparent to those skilled in the art from this detailed description.

The present invention provides a process for purification of high purity mogrosides with high mogroside V content from *Siraitia grosvenori* fruit extract.

Hereinafter the term "mogrosides" refers to mogrol and oxo-mogrol glycosides, including mogroside IIE, mogroside IIB, mogroside III, mogroside IV, mogroside V, 11-oxo-mogroside V, mogroside VI, siamenside I, and grosmomoside I.

Hereinafter the term "TM content" means the Total Mogrosides content, and it is calculated as the sum of 4 mogrosides including Mogroside V, 11-oxo-Mogroside V, Siamenside I, and Grosmomoside I.

Hereinafter the term "highly purified" or "high purity" means the TM content of at least 70% (w/w) on dry basis.

Hereinafter the term "high mogroside V content" means the mogroside V content of at least 60% (w/w) on dry basis.

Hereinafter the term "low purity" means the TM content of less than 70% (w/w) on dry basis.

Hereinafter the term "low mogroside V content" means mogroside V content of less than 60% (w/w) on dry basis.

Hereinafter the reported purity levels are determined by HPLC method with following parameters. HPLC system—"Agilent 1200 series" (USA) equipped with degasser, quaternary pump, autosampler, thermostatted column compartment and DAD UV detector. Column—"YMC Polyamine II" 4.6×150 mm 5 μm (Japan) at 30° C.; mobile phase 70:30 (vol/vol) acetonitrile water at 1 mL/min; detector—UV at 204 nm. Reference standards of mogroside V, 11-oxo-mogroside V, siamenside I, and grosmomoside I were obtained from "Chromadex Inc" (USA).

Hereinafter the term "impurity" means any compound other than mogrosides which are present in the mixture at more than 0.0001% (w/w) on dry basis. Non-limiting examples of impurities include proteins, pigments, polysaccharides, aldehydes, unsaturated aldehydes, methyl ketones, butyl crotonate, phenolic compounds as well as other non-mogroside compounds which may affect the organoleptic characteristics of Luo Han Guo sweetener.

The process of purification of mogrosides of the present invention is applicable for any low purity mogrosides mixture with the TM content of less than 70% w/w on dry basis.

The present invention also provides a column system that can be used for the process of purification of mogrosides of the present invention. The column system is with following parameters. The column system comprises a plurality of columns that are connected in either consecutive (serial) or parallel manner, where the plurality of columns are of same or different volumes packed with a sorbent with different affinity to impurities and steviol glycosides. When the column system is used in parallel connection mode, the inlet of each column may connect to a separate feed source while the outlet to a separate receiver. The following description will highlight the mode of column connection specific for each particular stage. Within the same stage the system may function as an entity of several parallel and serial connected column groups and separate columns. The number of columns is 3-15. The ratio of volumes of first column to volume of second column is 1:1 to 1:10. The ratio of volumes of last column to volume of previous column is 3:1 to 1:10. The columns are packed with sorbent up to 75-100% of their total volume. The columns are maintained at temperature 5-80° C.

The alcohol is selected from the group consisting of alkanols, more particularly methanol, ethanol, n-propanol, 2-propanol, 1-butanol, 2-butanol.

The sorbent is any macroporous polymeric adsorption resin capable of adsorbing mogrosides, such as Amberlite® XAD series (Rohm and Haas), Diaion® HP series (Mitsubishi Chemical Corp), Sepabeads® SP series (Mitsubishi Chemical Corp), Cangzhou Yuanwei YWD series (Cangzhou Yuanwei Chemical Co. Ltd., China) or equivalent.

The process of the present invention comprises a few stages. In one embodiment, the first stage comprises passing an aqueous or aqueous alcoholic solution of low purity mogrosides mixture through a consecutively connected column system. The water to alcohol ratio (vol/vol) in the aqueous alcoholic solution is 99.9:0.1 to 60:40. As a result, the impurities and different mogrosides are retained in different sections of the column system. Impurities with high affinity are mostly retained in the first column, and low affinity impurities are mostly retained in the last column, while mogrosides are mostly retained in the middle columns of the column system. Lower molecular weight mogrosides, particularly siamenside I and grosmomoside I, partially

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separated from higher molecular weight mogrosides, are retained at higher amounts in the first few columns of the middle columns of the column system, whereas higher molecular weight mogrosides, particularly mogroside V, are retained in the last few columns of the middle columns of the column system.

In one embodiment, the column system is composed of 6 columns that are connected in a consecutive manner. In the first stage, an aqueous alcoholic solution comprising a low purity mogrosides mixture with 34.73% (w/w on dry basis) comprising Mogroside V 19.47%, 11-oxo-Mogroside V 5.60%, Siamenoside I 5.20%, and Grosmomoside I 4.46% was passed through the column system; then each column was eluted with pure alcohol for total elution in order to check the retentions of mogrosides by each column. The results of the total elution from the first stage are summarized in Table 2.

TABLE 2

Mogrosides distribution after first stage							
Mogrosides, % w/w dry basis	Columns						
	Initial	I	II	III	IV	V	VI
Siamenoside I	5.20	2.90	11.30	8.60	4.10	2.81	1.52
Grosmomoside I	4.46	2.80	9.66	7.50	3.13	2.19	1.48
11-oxo-mogroside V	5.60	1.50	2.30	3.30	9.37	11.30	5.80
Mogroside V	19.47	2.73	2.33	17.94	40.32	44.54	8.96
TM content	34.73	9.93	25.59	37.34	56.92	60.84	17.76

In another embodiment of present invention, the second stage of the purification process comprises washing the columns with retained mogrosides from the first stage to remove residual impurities which still remain after the first stage of separation in order to increase further the purity level of final products. In one embodiment, the columns are sequentially washed with an acidic aqueous or aqueous alcoholic solution, a basic aqueous or aqueous alcoholic solution, and an aqueous alcoholic solution. The water to alcohol ratio (vol/vol) in aqueous alcoholic solution is 99.9:0.1 to 60:40. The removal of impurities is carried out either from each column separately (parallel connection) or more than one consecutively (serial) connected column groups. Similar to the first stage, each column was eluted with pure alcohol for total elution in order to check the retentions of mogrosides by each column. The results of the total elution from the second stage with the usage of the same low purity mogrosides mixture and the same 6 column system as in the first stage are summarized in Table 2.

TABLE 3

Mogrosides distribution after second stage							
Mogrosides, % w/w dry basis	Columns						
	Initial	I	II	III	IV	V	VI
Siamenoside I	5.20	5.28	20.41	14.77	6.26	4.03	2.56
Grosmomoside I	4.46	5.09	17.45	12.88	4.73	3.16	2.54
11-oxo-mogroside V	5.60	2.73	4.15	5.67	14.34	16.25	9.85
Mogroside V	19.47	4.96	4.21	30.81	61.52	64.05	15.22
TM content	34.73	18.06	46.23	64.13	86.85	87.49	30.16

In one embodiment of present invention, the third stage of the purification process comprises eluting the middle section columns (e.g., columns IV and V) with an aqueous alcoholic solution, where water to alcohol ratio (vol/vol) of the aqueous alcoholic solution is 60:40 to 0.1:99.9. The elution

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is carried out either from each column separately (parallel connection) or more than one consecutively (serial) connected column groups. The elution results in high purity mogrosides.

In one embodiment, the fourth stage of the purification process comprises removing alcohol from mogrosides eluate obtained after the third stage and further concentrating and drying the mogrosides eluate to obtain a dried highly purified mixture of mogrosides. Any method known to art may be used for ethanol removal, concentration and drying.

In one embodiment, the low purity fractions are combined and subjected to repeated purification from the first stage to the fourth stage as described above.

In one embodiment, the high purity fractions are combined and dried to produce high mogroside V content highly purified mixture of mogrosides having TM content of 87.18% (w/w on dry basis) comprising Mogroside V 62.82%, 11-oxo-Mogroside V 15.32%, Siamenoside I 5.11%, and Grosmomoside I 3.92%.

In one embodiment, the purification process of the present invention further comprises filtering with the usage of ultrafiltration and/or nanofiltration membranes. Membranes with molecular weight cut-off (MWCO) size of 1000, 1500 and 2000 are used. Highly purified mogroside mixture comprising Mogroside V 62.82%, 11-oxo-Mogroside V 15.32%, Siamenoside I 5.11%, and Grosmomoside I 3.92% was dissolved in aqueous alcohol solution with alcohol content of 0-100% (vol/vol) preferably 0-50%, to make a solution with total solids content of 0.1-50% (wt/vol), preferably 0.5-10%. The resulted solution was consecutively passed through ultrafiltration and/or nanofiltration membranes with MWCO 1000, 1500, 2000 and 2500. A stirred cell membrane system from Sterlitech Corp. (USA) was used for this purpose. Anyway any suitable filtration system known to art may be used for this purpose. Non-limiting examples of membrane manufacturers are Koch Membrane Systems Inc. (USA), GE-Osmonics (USA), Alfa Laval (Sweden). Flat sheet, hollow fiber, spiral and other membranes may be used. Diafiltration was used to increase membrane filtration process efficiency. Depending on membrane size the retentate or permeate contained the main amount of mogrosides. After each membrane treatment the mogroside containing fraction (retentate or permeate) was concentrated or diluted again till total solids content 0.1-50% (wt/vol) preferably 0.5-10% and passed through the next membrane. The solution was passed through increasing membrane sizes (from MWCO 1000 to 2500). Anyway other order of membrane is also possible. After membrane treatment highly purified mogroside mixture was obtained comprising Mogroside V 67.13%, 11-oxo-Mogroside V 16.31%, Siamenoside I 5.51%, and Grosmomoside I 4.10% (dry basis).

The high purity mogrosides can be used either alone or in combination with other high intensity sweeteners in food, beverage, pharmaceutical composition, tobacco, nutraceutical, oral hygienic composition, or cosmetic. The other high intensity sweeteners include steviol glycosides including a purified sweet steviol glycoside mixture, stevioside, rebaudioside A, rebaudioside B, rebaudioside C, rebaudioside D, rebaudioside E, rebaudioside F, dulcoside A, dulcoside B, rubusoside, and stevia; siamenoside; mogroside IV; mogroside V; Luo Han Guo sweetener; monatin and its salts (monatin SS, RR, RS, SR); glycyrrhizic acid and its salts; curculin; thaumatin; monellin; mabinlin; brazzein; hennandulcin; phyllodulcin; glycyphyllin; phloridzin; trilobatin; baiyunoside; osladin; polypodoside A; pterocaryoside A;

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pterocaryoside B; mukurozioside; phlomiside I; perian-
drin I; abrusoside A; cyclocarioside I; and combinations
thereof.

The following examples are provided for illustrating the
process and column system of the present invention.

Example 1

Purification of High Purity Mogrosides

100 g of Luo Han Guo extract (w/w on dry basis) compris-
ing Mogroside V 19.47%, 11-oxo-Mogroside V 5.60%,
Siamenoside I 5.20%, and Grosmomomide I 4.46% (TM
content 34.73%) was dissolved in 2.4 liters of deionized
water. The solution was passed through 6 consecutively
connected 200 mL columns packed with Diaion® HP (Mit-
subishi Chemical Corp) macroporous adsorbent. 2.4 L
deionized water was subsequently passed through the sys-
tem. Further the columns were separately washed with 2
volumes of 0.5% HCl, then with water until neutral pH, then
with 0.5% NaOH and again with water until neutral pH of
washing waters is achieved. Then the columns were sepa-
rately washed with 10% Ethanol. Then the columns were
eluted with 50% Ethanol. Columns 4 and 5 were connected
consecutively. The eluate of column 4 and 5 was evaporated
under vacuum for ethanol removal. Obtained aqueous solu-
tion was spray dried to yield 31 g of highly purified
mogroside mixture comprising Mogroside V 62.82%,
11-oxo-Mogroside V 15.32%, Siamenoside I 5.11%, and
Grosmomomide I 3.92% (on dry basis).

Example 2

Membrane Purification.

100 g of high purity mogrosides prepared according to
EXAMPLE 1 comprising Mogroside V 62.82%, 11-oxo-
Mogroside V 15.32%, Siamenoside I 5.11%, and Grosmo-
momide I 3.92% (on dry basis) was dissolved in 2.4 liters of
deionized water. The obtained solution was further passed
through stirred cell membrane system (Sterlitech Corp.,
USA). The solution was passed through GE Osmonics 2500
MWCO membrane. The permeate which contained the
mogrosides was concentrated to 5% solids and further
passed through 1000MWCO membrane. The retentate
which contained the mogrosides was concentrate and dried
to yield 90 g powder containing Mogroside V 67.13%,

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11-oxo-Mogroside V 16.31%, Siamenoside I 5.51%, and
Grosmomomide I 4.10% (dry basis).

While the present invention has been described with
reference to particular embodiments, it will be understood
that the embodiments are illustrative and that the invention
scope is not so limited. Alternative embodiments of the
present invention will become apparent to those having
ordinary skill in the art to which the present invention
pertains. Such alternate embodiments are considered to be
encompassed within the spirit and scope of the present
invention. Accordingly, the scope of the present invention is
described by the appended claims and is supported by the
foregoing description.

I claim:

1. A high purity mogrosides mixture comprising:

Mogroside V in the range of 60-70wt %;

11-oxo-Mogroside V in the range of 10-20wt %;

Siamenoside I in the range of 3-7wt %; and

Grosmomomide I in the range of 2-6wt %.

2. A sweetener mixture comprising the high purity mogro-
sides mixture of claim 1, wherein the high purity mogrosides
mixture is blended with another high intensity sweetener.

3. The sweetener mixture comprising the high purity
mogrosides mixture of claim 2, wherein the another high
intensity sweetener is selected from the group consisting of
steviol glycosides including a purified sweet steviol glyco-
side mixture, stevioside, rebaudioside A, rebaudioside B,
rebaudioside C, rebaudioside D, rebaudioside E, rebaudio-
side F, dulcoside A, dulcoside B, rubusoside, and stevia;
siamenoside; mogroside IV; mogroside V; Luo Han Guo
sweetener; monatin and its salts (monatin SS, RR, RS, SR);
glycyrrhizic acid and its salts; curculin;

thaumatin; monellin; mabinlin; brazzein; hernandulcin;
phyllodulcin; glycyphyllin; phloridzin;

trilobatin; baiyunoside; osladin; polypodoside A; ptero-
caryoside A; pterocaryoside B;

mukurozioside; phlomiside I; perian-
drin I; abrusoside A; cyclocarioside I; and combinations thereof.

4. A product comprising the high purity mogrosides
mixture of claim 2.

5. The product of claim 4, wherein the product is selected
from the group consisting of food, beverage, pharmaceutical
composition, tobacco, nutraceutical, oral hygienic compo-
sition, or cosmetic.

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